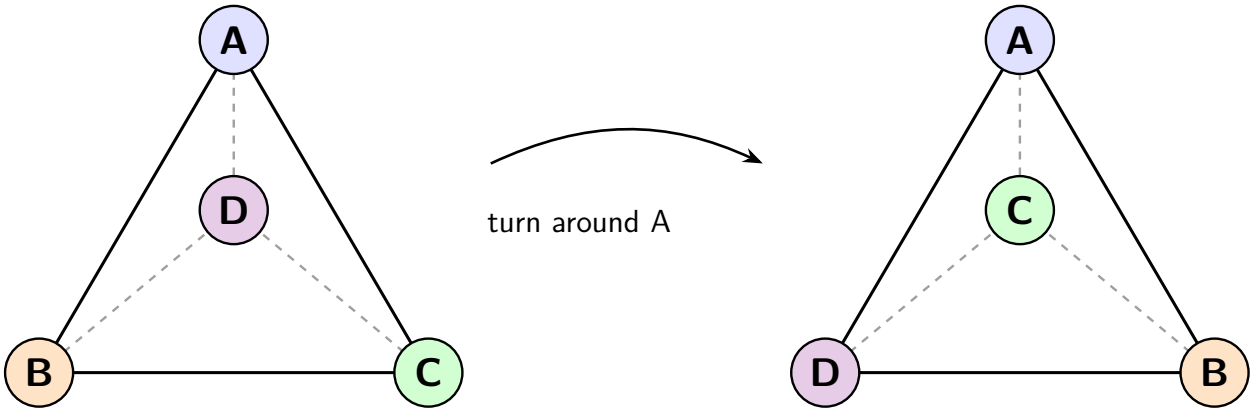
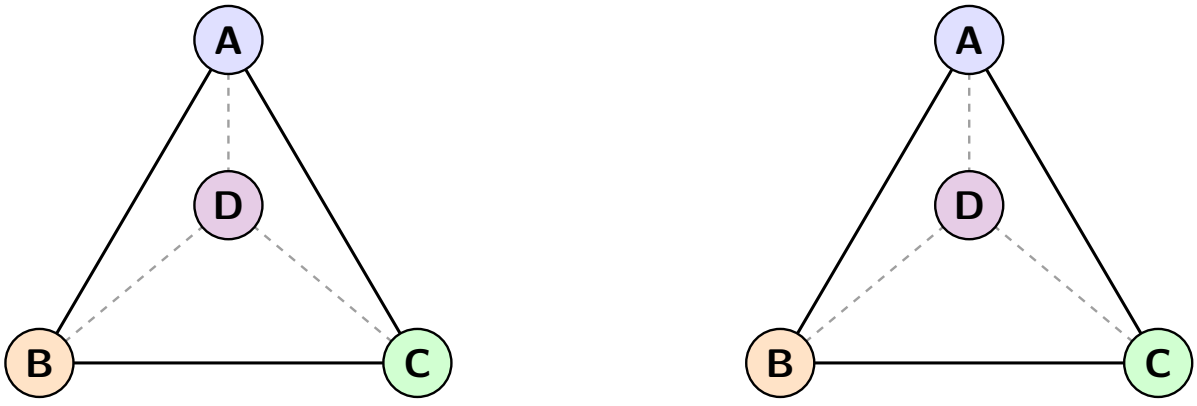
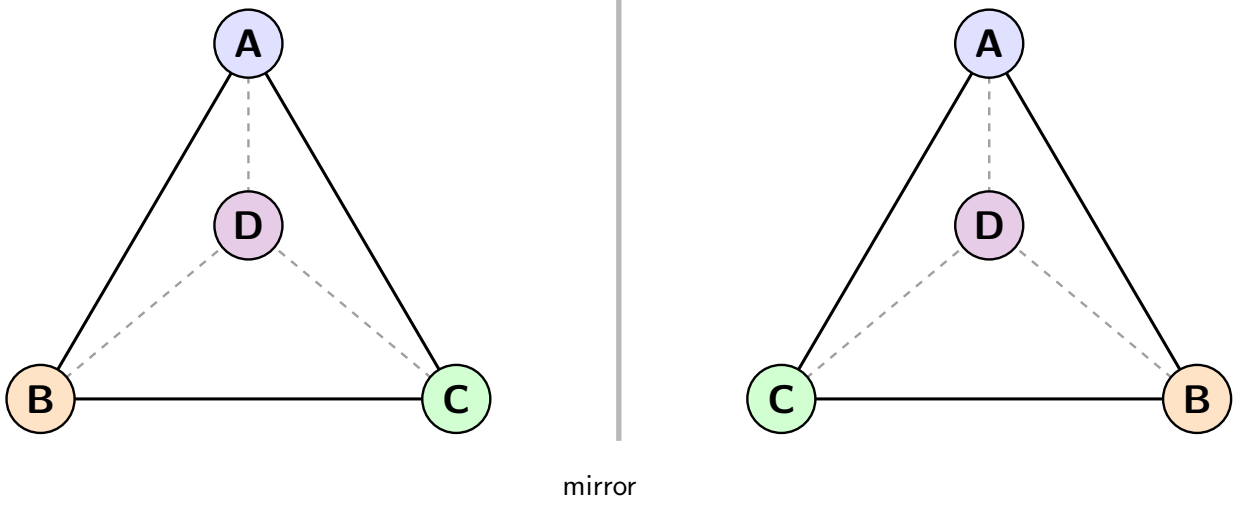
Use the rigid models. Move and turn them so same-letter corners match. Which way a letter points does not count. Keep the markers fixed unless a problem says to change them. Dashed edges pass behind the front face.



Problem 1: Build two copies of the first model. Take turns hiding and rotating one model, then matching all four letters again.

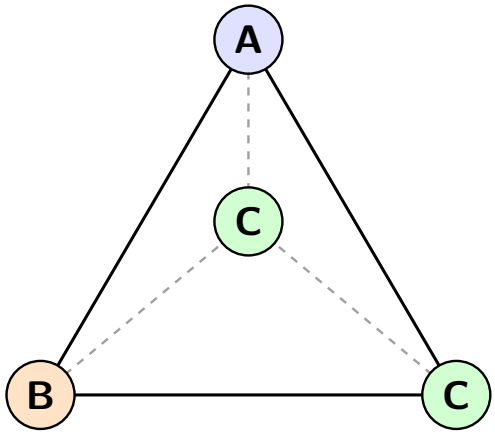


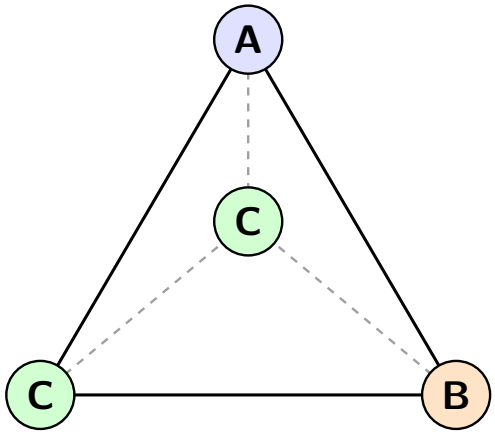
Problem 2: Build this model and its mirror twin. Can you match every letter by moving and turning one of them? If no turn has worked, does that mean no turn can work?



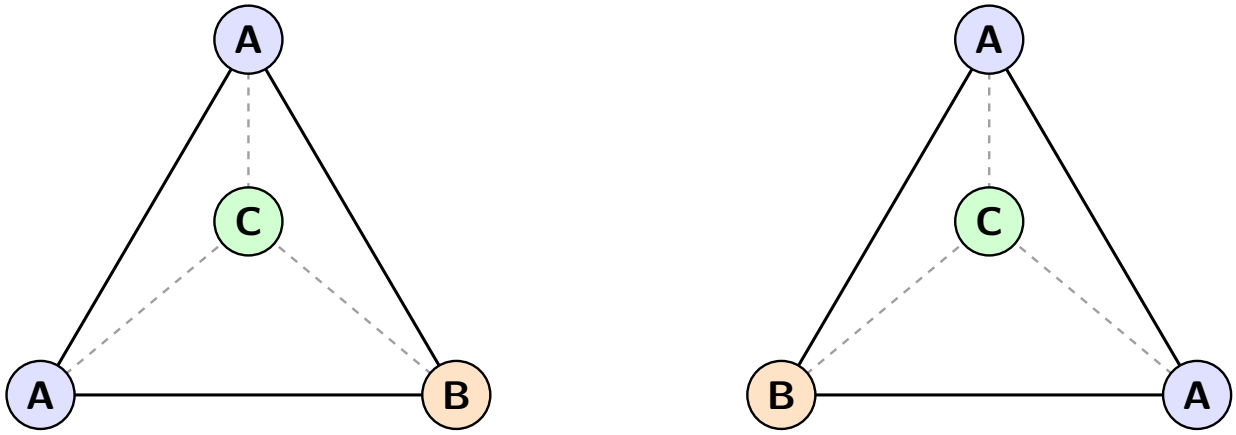
Problem 3: Keep the pair from Problem 2. Put A, B, C and D on another model in any way you like. Which of the two does it match? Can you make one that matches neither?

Problem 4: Change D to C on both models from Problem 2 and test for a match. Try other choices of two letters to make equal; start each trial with the original A, B, C, D mirror pair. Which choices allow a match by turning?





Problem 5: Use A, A, B and C on each model. Find out whether your partner can make a pair that will not match by turning.



Problem 6: Change one A on each model to D. Can you choose the changes to make a matching pair? Can you choose them to make a pair that will not match?